Financial Accounting

BC-2

31th March 2021

m.asif@fuuast.edu.pk

- **Financial**
  - The finances or financial situation of an organization or individual. In term of monetary

- **Accounting**
  - Accounting is the process of recording financial transactions pertaining to a business.
    The accounting process includes summarizing, analyzing and reporting these transactions to oversight agencies, regulators and tax collection entities.

- **Accounting**
  - Main Goal & Objective is <u>" To provide the Information"</u>
    - **Information**
      - **Financial Information**
        - » **Income Statement**
        - » **Balance Sheet**
        - » **Cash flow Statement**
        - » **Change in equity**
      - **Managerial (Insider)**

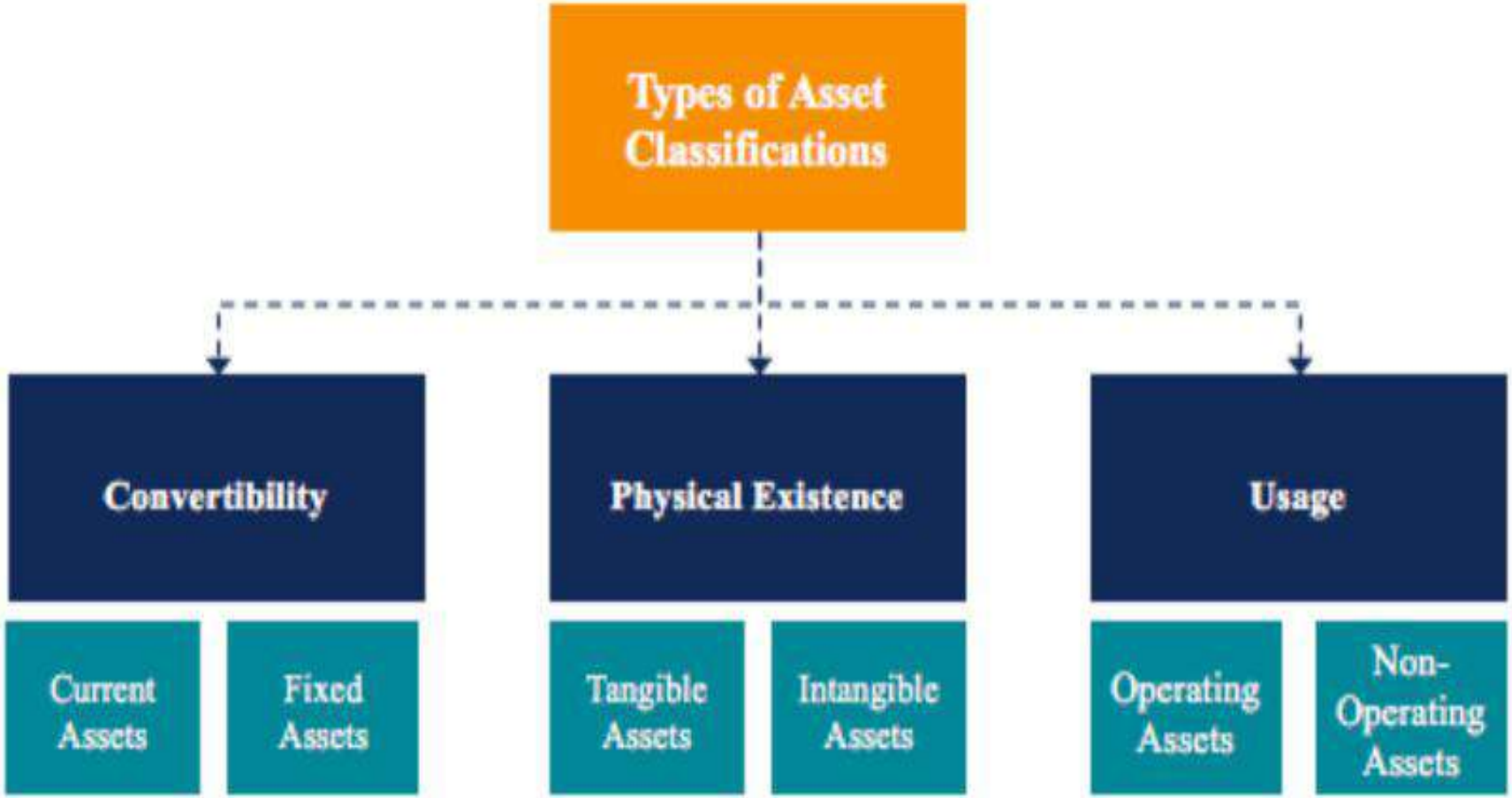
Outsider **Banker , Investor, Creditor Govt.Agency**

Financial Accounting

Accounting Equation

Accounting Equation: Assets = Liabilities + Owner's Equity

- **Assets**
  - An asset is a resource with economic value that an individual, corporation, or country owns or controls with the expectation that it will provide a future benefit. Assets are reported on a company's balance sheet and are bought or created to increase a firm's value or benefit the firm's operations.


Types of Asset Classifications
Convertibility
Physical Existence
Usage
Current Assets
Fixed Assets
Tangible Assets
Intangible Assets
Operating Assets
Non-Operating Assets

- **1. Current Assets**
  - Current assets are assets that can be easily converted into cash and cash equivalents (typically within a year). Current assets are also termed liquid assets and examples of such are:
    - Cash
    - Cash equivalents
    - Short-term deposits
    - Accounts receivables
    - Inventory
    - Marketable securities
    - Office supplies

- **2. Fixed or Non-Current Assets**
- Non-current assets are assets that cannot be easily and readily converted into cash and cash equivalents. Non-current assets are also termed fixed assets, long-term assets, or hard assets. Examples of non-current or fixed assets include:
- Land
- Building
- Machinery
- Equipment
- Patents
- Trademarks

- **Classification of Assets: Physical Existence**
  - If assets are classified based on their physical existence, assets are classified as either **tangible assets** or **intangible assets**.
  - **1. Tangible Assets**
    - Tangible assets are assets with physical existence (we can touch, feel, and see them). Examples of tangible assets include:
    - Land
    - Building
    - Machinery
    - Equipment
    - Cash
    - Office supplies
    - Inventory
    - Marketable securities

- **2. Intangible Assets**
  - Intangible assets are assets that lack physical existence. Examples of intangible assets include:
    - Goodwill
    - Patents
    - Brand
    - Copyrights
    - Trademarks
    - Trade secrets
    - Licenses and permits
    - Corporate intellectual property

- **Classification of Assets: Usage**
  - If assets are classified based on their usage or purpose, assets are classified as either **operating assets** or **non-operating assets.**
- **1. Operating Assets**
  - Operating assets are assets that are required in the daily operation of a business. In other words, operating assets are used to generate revenue from a company's core business activities. Examples of operating assets include:
    - Cash
    - Accounts receivable
    - Inventory
    - Building
    - Machinery
    - Equipment
    - Patents
    - Copyrights
    - Goodwill

- **2. Non-Operating Assets**
  - Non-operating assets are assets that are not required for daily business operations but can still generate revenue. Examples of non-operating assets include:
    - Short-term investments
    - Marketable securities
    - Vacant land
    - Interest income from a fixed deposit

Financial Accounting

Terminologies

Assets

Current assets

i) Cash at bank
ii) Cash on hand
iii) Stock of goods (purchase for re-sale)
iv) Investment
v) Prepaid
vi) Debtor (Credit Sales)
vii) Note receivable
viii) Office Supplies
ix) Accrued Income

Fixed Assets

i) Premises
ii) Motor vehicles
iii) Office furniture
iv) Office machines
v) Plant
vi) Equipment
vii) Land
etc

- **Liability**
  - A **liability** (Payable) is something a person or company owes, usually a sum of money

Financial Accounting

Terminologies

Liabilities: Liabilities are amount owed by a business to external parties.

Liabilities

Current Liabilities
(with in one year)

i) Creditor (Credit Purchases)
ii) Accrued Salaries
iii) Interest Outstanding
iv) Rent payable
v) Bank overdraft
etc

Long-term Liabilities
(Over one year)

i) Loan
ii) Debentures
iii) Bonds
iv) TFCs

- **Owner's Equity**
  - **Owner's Equity** is defined as the proportion of the total value of a company's assets that can be claimed by its **owners**

Financial Accounting

Terminologies

Owner's Equity: Owner's Equity is the Owner's interest or claim on the assets of the business.

Owner's Equity

i) Investment +

ii) Profit +

iii) Loss -

iv) Withdrawals -

Financial Accounting

Accounting Equation

- Assets = Liabilities + Owner Equity

  100,000 = 30,000 + 70,000

- Assets = Liabilities + Owner Equity(capital+P/L)

- Profit/Loss = Revenue – Expense

Financial Accounting

Accounting Equation Q1

1.1 Complete the gaps in the following table:

	Assets	*Liabilities*	*Capital*
	£	£	£
(*a*)	12,500	1,800	?
(*b*)	28,000	4,900	?
(*c*)	16,800	?	12,500
(*d*)	19,600	?	16,450
(*e*)	?	6,300	19,200
(*f*)	?	11,650	39,750

1.2A Complete the gaps in the following table:

	Assets	*Liabilities*	*Capital*
	£	£	£
(*a*)	55,000	16,900	?
(*b*)	?	17,200	34,400
(*c*)	36,100	?	28,500
(*d*)	119,500	15,400	?
(*e*)	88,000	?	62,000
(*f*)	?	49,000	110,000

Financial Accounting

- **Types of Organization**

Business *(Trading, Manufacturing, Service)*

Non Business *(NGOs)*

Financial Accounting

- **Types of business Unit / Organization**

1. **Sole Proprietorship**
2. **Partnership**
3. **Public Ltd. Companies**

Financial Accounting

- **Types of Revenues**

Revenues can be classified as operating revenue and non-operating revenue.

1. **Operating revenues** are those that originate from main business operations. For example: **Sales**, etc.
2. **Non-operating revenues** are earned from some side activity. For example: **Interest Revenue, Rent Revenue, Commission Revenue**(except in case where the business' main industry is renting industry).

Financial Accounting

- **Expenses definition**.

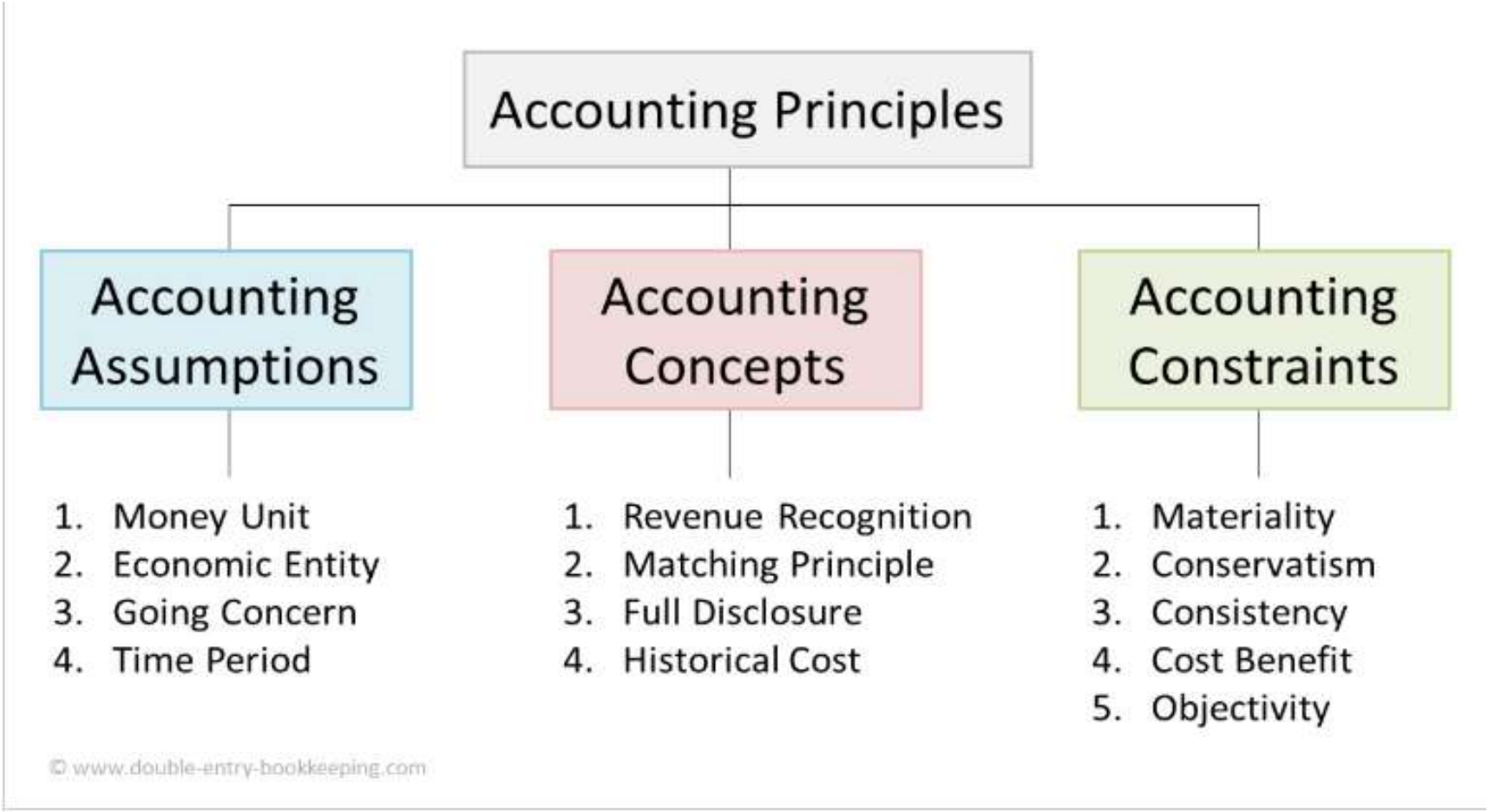
  Costs that are matched with revenues on the income statement. For example, Cost of Goods Sold is an expense caused by Sales. Insurance Expense, Wages Expense, Advertising Expense,Interest Expense are expe nses matched with the period of time in the heading of the income statement.

Financial Accounting

- Following is a list of common **types of expenses** recognized in the financial statements:
- Salaries and wages
- Utility expenses
- Cost of goods sold
- Administration expenses
- Finance costs (Interest Exp)
- Depreciation (Non Cash)
- Rent Etc

Financial Accounting

Accounting Principles



Financial Accounting

Accounting Principles

- **MONEY MEASUREMENT**

All transactions of the business are recorded in terms of money

- **Economic Entity**

1. Sole Proprietorship
2. Partnership
3. Ltd. Companies

Financial Accounting

Accounting Principles

- **Going concern concept**

The business will continue in operational existence for the foreseeable future

- **Time Period**

is an accounting principle which states that a business should report their financial statements appropriate to a specific time period.

Financial Accounting

Accounting Principles

- **Revenue recognition principle**,

a feature of accrual accounting, requires that revenues are recognized on the income statement in the period when realized and earned—<u>not necessarily when cash is received</u>

- **Matching principle** states that expenses should be recognized and recorded when those expenses can be matched with the revenues those expenses helped to generate.

Financial Accounting

Accounting Principles

- **Full disclosure principle**

is a concept that requires a business to report all necessary information about their financial statements and other relevant information to any persons who are accustomed to reading this information.

- **Historical Cost Principle**.

states that businesses must record and account for most assets and liabilities at their purchase or acquisition price. In other words, businesses have to record an asset on their balance sheet for the amount paid for the asset.

Financial Accounting

Accounting Principles

- **Materiality principle**

states that an accounting standard can be ignored if the net impact of doing so has such a small impact on the financial statements that a user of the statements would not be misled.

Financial Accounting

Accounting Principles

- **Conservatism principle** is the general concept of recognizing expenses and liabilities as soon as possible when there is uncertainty about the outcome, but to only recognize revenues and assets when they are assured of being received.

Financial Accounting

Accounting Principles

Consistency

refers to the principle that companies should use the same accounting methods to record similar transactions over time. In other words, companies shouldn't bounce between accounting rules and treatments to manipulate profits or other financial statement elements.

Financial Accounting

Accounting Principles

- **Cost benefit principle**

relationship states that the cost of providing financial information in the financial statements must not outweigh the benefit of that information to the users. In other words, financial information is not free.

Financial Accounting

Accounting Principles

- **Objectivity** is a philosophical **concept** of being true independently from individual subjectivity caused by perception, emotions, or imagination. A proposition is considered to have objective truth when its truth conditions are met without bias caused by a sentient subject.

Financial Accounting

Thank you

Financial & Cost accounting

Financial & Cost accounting

- **Definition of Cost Accounting**
  - **Cost Accounting** is a branch of accounting that focuses on **capturing, analyzing, and managing the costs** associated with producing goods or services. It helps businesses:
    - **Determine product/service costs** accurately.
    - **Control expenses** and improve efficiency.
    - **Support decision-making** (pricing, budgeting, profitability analysis).
    - **Compare actual costs with budgets** (variance analysis).
- **Key Features:**
  - **Internal Focus**: Used by management (unlike financial accounting, which targets external stakeholders).
  - **Cost Classification**: Segregates costs into **direct/indirect, fixed/variable, manufacturing/non-manufacturing**.
  - **Methods**: Includes **job costing, process costing, activity-based costing (ABC), standard costing**.
  - **Tools**: Break-even analysis, marginal costing, cost-volume-profit (CVP) analysis.

Financial & Cost accounting

Difference from Financial Accounting:

Aspect	Cost Accounting	Financial Accounting
Purpose	Internal decision-making	External reporting (investors, regulators)
Reporting Time	Flexible (as needed	Fixed (quarterly/annually)
Standards	No strict rules	(company-specific) Follows GAAP/IFRS

Financial & Cost accounting

Cost Classifications

Behavior Fixed, variable, and mixed costs

direct vs. indirect costs

cost behavior analysis.

Financial & Cost accounting

- **Introduction to Cost Classifications**
  - **Definition of Costs**: A cost is an economic sacrifice made to achieve a business goal, typically related to production or service provision.
  - **Purpose of Cost Classification**: Classifying costs helps in budgeting, decision-making, and controlling operations. It also aids in determining profitability and pricing decisions.

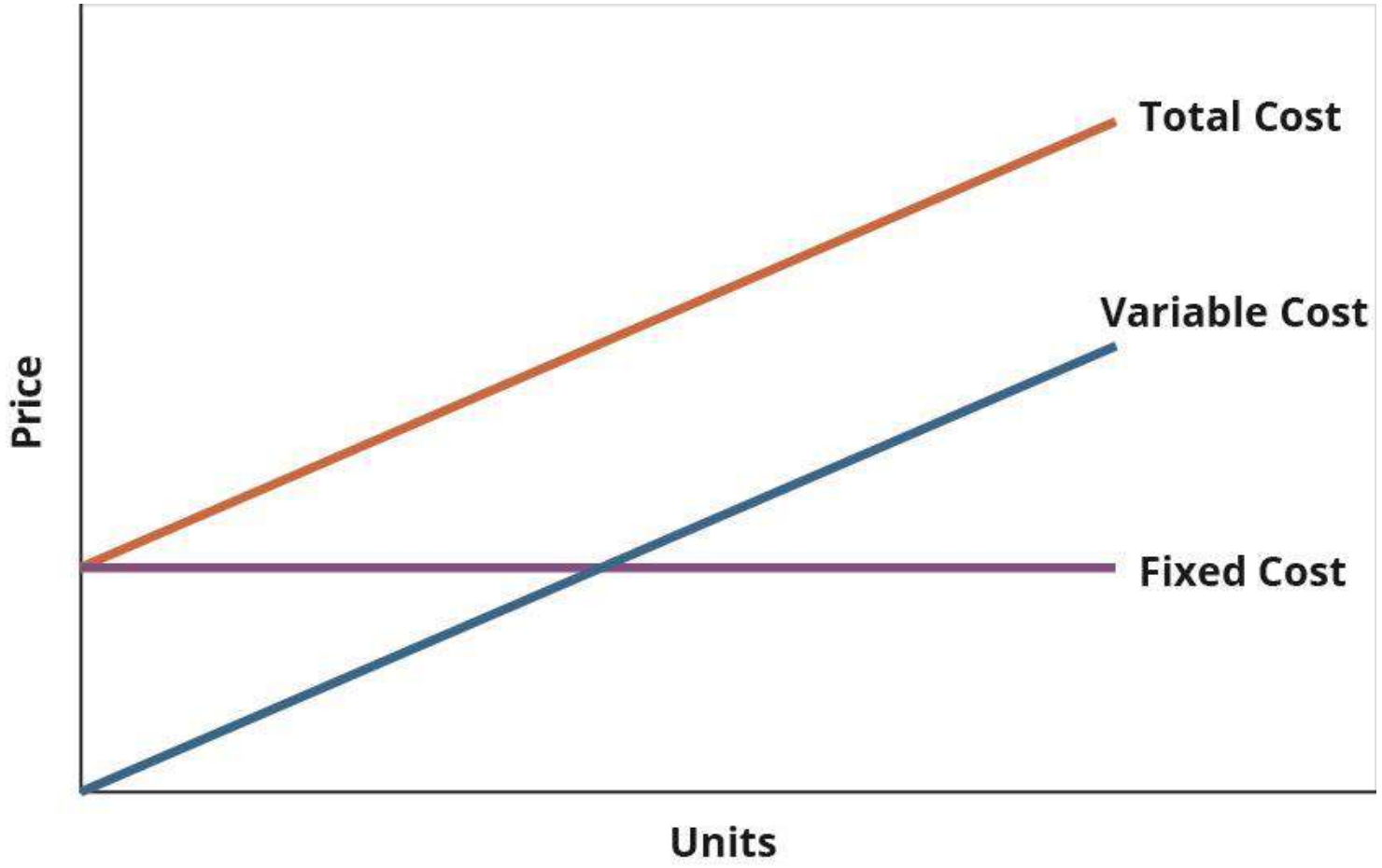
Financial & Cost accounting

- **A) Fixed, Variable, and Mixed Costs**
  - **Fixed Costs**
    - **Definition**: Costs that remain constant, regardless of the level of production or business activity within a relevant range.
    - **Examples**:
      - Factory rent
      - Salaries of permanent staff
      - Insurance premiums
  - **Characteristics**:
    - Do not change with production volume in the short term.
    - Become smaller per unit as production increases (due to fixed cost spreading).

Financial & Cost accounting

- **B) Variable Costs**
  - **Definition**: Costs that vary directly with the level of production or business activity.
  - **Examples**:
    - Direct materials (e.g., fabric in garment manufacturing)
    - Direct labor (e.g., wages for hourly workers)
    - Utility costs tied to production volume
  - **Characteristics**:
    - Change proportionally with production levels.
    - Total variable cost increases with output, but per-unit variable cost remains constant.
  - **Graph**: Show a linear relationship between total variable costs and production levels.

Financial & Cost accounting



Financial & Cost accounting

- **C) Mixed Costs**
  - **Definition**: Costs that have both fixed and variable components.
  - **Examples**:
    - Electricity bills (a fixed monthly charge + additional costs based on usage)
    - Salesperson’s salary (base salary + commission)
  - **Characteristics**:
    - Part of the cost remains fixed, while the other part changes with activity level.
- **Conversion Costs**
  - The total conversion cost formula is: Conversion Costs = Direct Labor Costs + Manufacturing Overheads

Financial & Cost accounting

Direct materials	**Prime cost**	**Production cost**	**Total cost**
Direct labour			
Direct expenses			
Plus			
Indirect manufacturing costs			
Plus			
Administration expenses			
Selling and distribution expenses			
Financial charges			

Financial & Cost accounting

- **Inventories**
  - **Raw Material Beginning inventories**
  - **Raw Material Ending inventories**
  - **Work in Process Beginning inventories**
  - **Work in Process Ending inventories**
  - **Finished goods Beginning inventories**
  - **Finished goods Ending inventories**

AKU & Co			
Manfacturing Statement			
31-Dec-23			
Raw material			
Beg. Inventory Material			10,000
Add: Raw material Purchase		500,000	
Add: Transportation In		15,000	
Total Purchase		515,000	
less: Purchase return		-2,000	
less: Purchase Discount		-3,000	
Net purchase			510,000
Material available for used			520,000
Less:Closing inventory			-35,000
Raw material consumed			485,000
Direct Labour			1,000,000
Direct Exp			25,000
Prime Cost			1,510,000
Factory overhead			
Indirect material		50,000	
Indirect Labour		500,000	
Factory Rent		700,000	
Factory utility		150,000	
Plant & Machinery Depreciation		135,000	
Factory Insurance		50,000	
Total Factory Over head			1,585,000
Total Manufacturing cost			3,095,000
Add: Work in process(Begg)			245,000
Less: Work in process(Ending)			-450,000
Total Manufacturing cost			**2,890,000**

AKU & Co		
Income Statement		
31-Dec-23		
Revenue		
Sales		5,000,000
less: Sales Return		-5,000
Net sales		4,995,000
Cost of goods sold		
Finish goods Beginning. Inv	500,000	
Total Manufacturing Cost	**2,890,000**	
Cost of goods available for sale	3,390,000	
less: Finish goods Ending. Inv	-390,000	
Cost of goods sold		3,000,000
Gross Profit		1,995,000
Less: Opreating Exp		
Selling & distribution Exp		895,000
Admin Exp		100,000
EBIT		1,000,000
less: Interest		150,000
EBT		850,000
less: Tax 40%		340000
Net profit after tax		510,000

Financial & Cost accounting

Thank you

Managerial Accounting

Thirteenth Edition

Ray H. Garrison, D.B.A., CPA

Professor Emeritus

Brigham Young University

Eric W. Noreen, Ph.D., CMA

Professor Emeritus

University of Washington

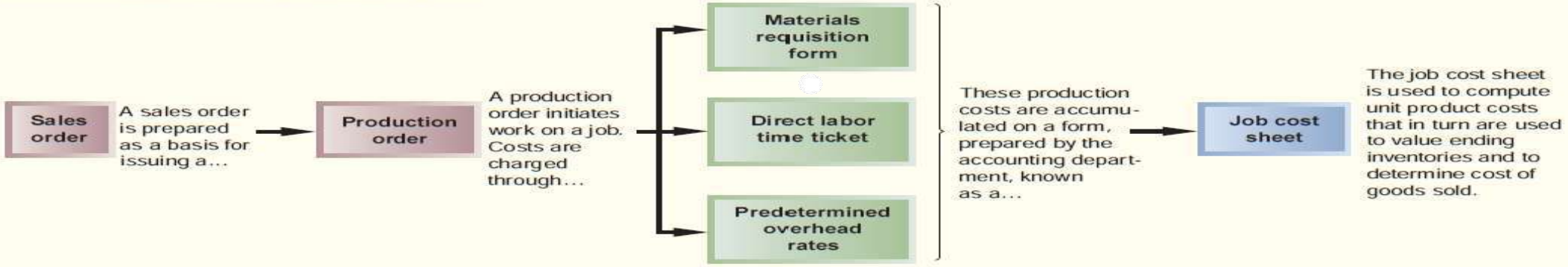
Peter C. Brewer, Ph.D., CPA

Miami University—Oxford, Ohio

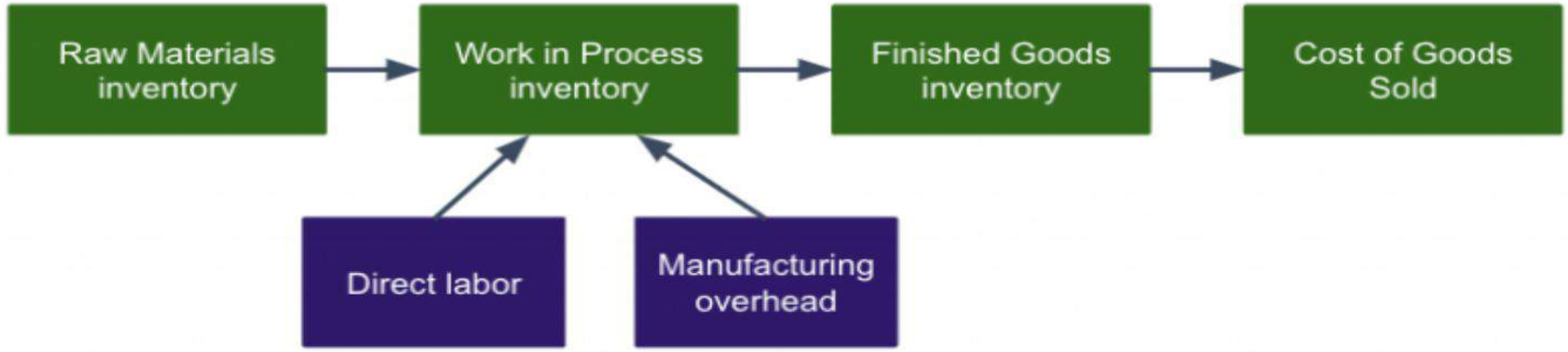
Boston

Job order Costing

EXHIBIT 3–5
The Flow of Documents in a Job-Order Costing System



Job costing



The Purchase and Issue of Materials

On April 1, Ruger Corporation had $7,000 in raw materials on hand. During the month, the company purchased on account an additional $60,000 in raw materials. The purchase is recorded in journal entry (1) below:

(1)

Raw Materials	60,000	
Accounts Payable		60,000

Issue of Direct and Indirect Materials During April, $52,000 in raw materials were requisitioned from the storeroom for use in production. These raw materials included $50,000 of direct and $2,000 of indirect materials. Entry (2) records issuing the materials to the production departments.

(2)

Work in Process	50,000	
Manufacturing Overhead	2,000	
Raw Materials		52,000

costs are classified as either direct or indirect labor. In April, $60,000 was recorded for direct labor and $15,000 for indirect labor. The following entry summarizes those events:

(3)

Work in Process	60,000	
Manufacturing Overhead	15,000	
Salaries and Wages Payable		75,000

Manufacturing Overhead Costs

Recall that all manufacturing costs other than direct materials and direct labor are classified as manufacturing overhead costs. These costs are entered directly into the Manufacturing Overhead account as they are incurred. To illustrate, assume that Ruger Corporation incurred the following general factory costs during April:

Utilities (heat, water, and power)	$21,000
Rent on factory equipment	16,000
Miscellaneous factory overhead costs	3,000
Total	$40,000

The following entry records the incurrence of these costs:

(4)

Manufacturing Overhead	40,000	
Accounts Payable*		40,000

*Accounts such as Cash may also be credited

In addition, assume that during April, Ruger Corporation recognized $13,000 in accrued property taxes and that $7,000 in prepaid insurance expired on factory buildings and equipment. The following entry records these items:

(5)

Manufacturing Overhead	20,000	
Property Taxes Payable		13,000
Prepaid Insurance		7,000

Finally, assume that the company recognized $18,000 in depreciation on factory equipment during April. The following entry records the accrual of this depreciation:

(6)

Manufacturing Overhead	18,000	
Accumulated Depreciation		18,000

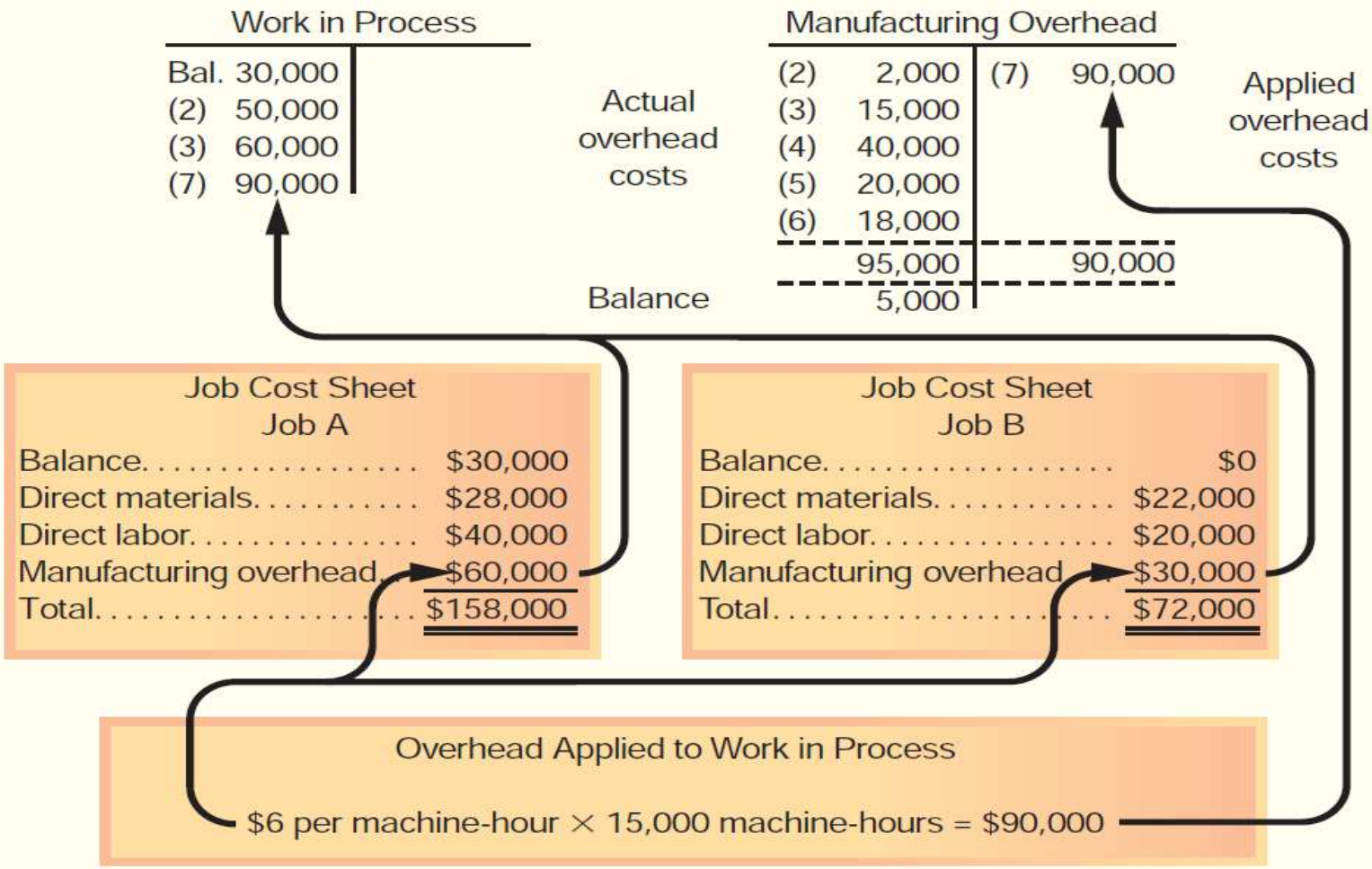
charged to the job.

To illustrate, assume that Ruger Corporation's predetermined overhead rate is $6 per machine-hour. Also assume that during April, 10,000 machine-hours were worked on Job A and 5,000 machine-hours were worked on Job B (a total of 15,000 machine-hours). Thus, $90,000 in overhead cost ($6 per machine-hour × 15,000 machine-hours = $90,000) would be applied to Work in Process. The following entry records the application of Manufacturing Overhead to Work in Process:

(7)

Work in Process	90,000	
Manufacturing Overhead		90,000

EXHIBIT 3–8
The Flow of Costs in Overhead Application



In the case of Ruger Corporation, assume that Job A was completed during April. The following entry transfers the cost of Job A from Work in Process to Finished Goods:

(11)

Finished Goods	158,000	
Work in Process		158,000

The $158,000 represents the completed cost of Job A, as shown on the job cost sheet in Exhibit 3–8. Because Job A was the only job completed during April, the $158,000 also represents the cost of goods manufactured for the month.

For Ruger Corporation, we will assume 750 of the 1,000 gold medallions in Job A were shipped to customers by the end of the month for total sales revenue of $225,000. Because 1,000 units were produced and the total cost of the job from the job cost sheet was $158,000, the unit product cost was $158. The following journal entries would record the sale (all sales were on account):

(12)

Accounts Receivable	225,000	
Sales		225,000

(13)

Cost of Goods Sold	118,500	
Finished Goods		118,500
(750 units × $158 per unit = $118,500)		

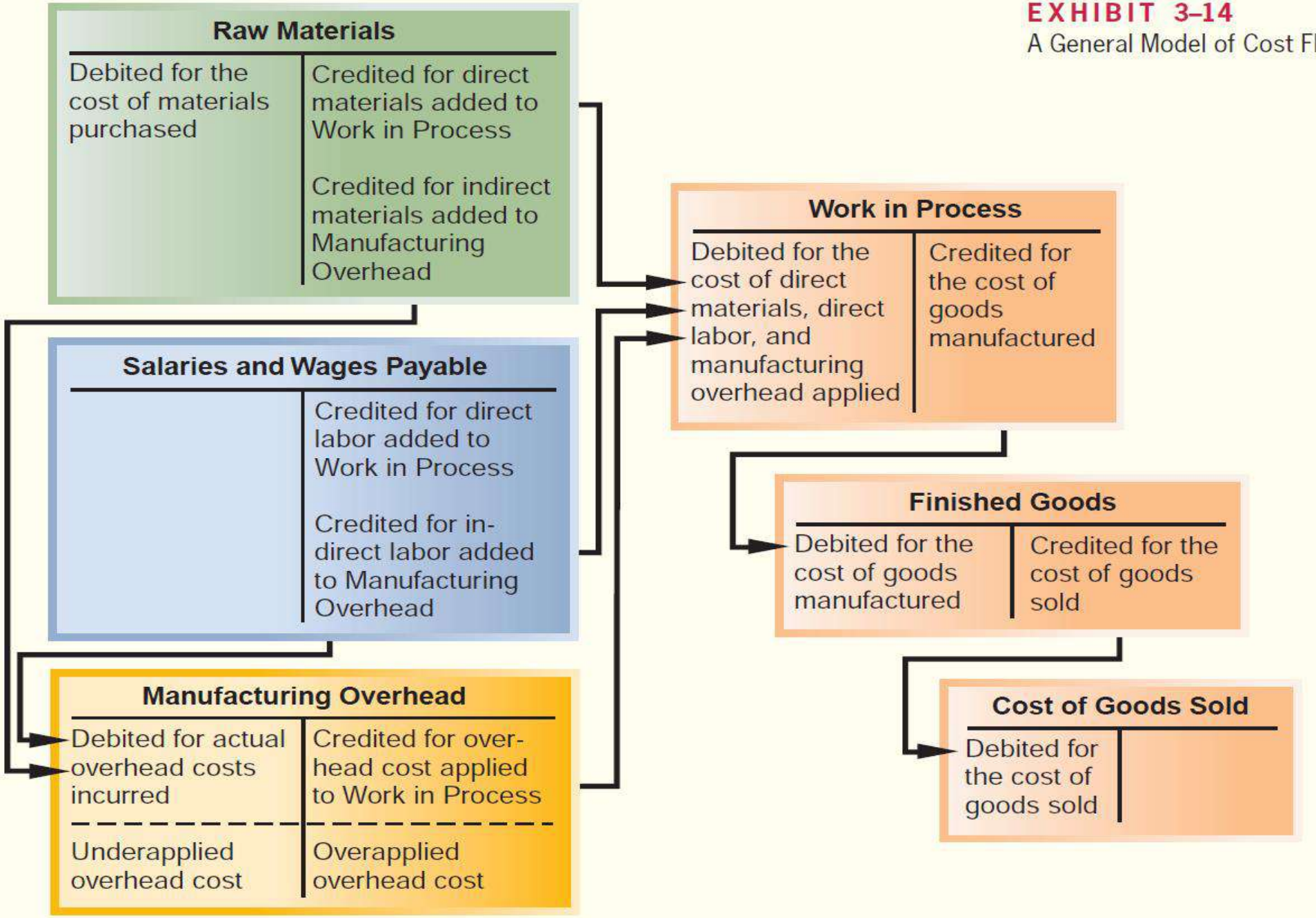
Closed Out to Cost of Goods Sold Closing out the balance in Manufacturing Overhead to Cost of Goods Sold is simpler than the allocation method. Returning to the example of Ruger Corporation, the entry to close the $5,000 of underapplied overhead to Cost of Goods Sold is:

(14)		
Cost of Goods Sold	5,000	
Manufacturing Overhead		5,000

Note that because the Manufacturing Overhead account has a debit balance, Manufacturing Overhead must be credited to close out the account. This has the effect of increasing Cost of Goods Sold for April to $123,500:

Unadjusted cost of goods sold [from entry (13)]	$118,500
Add underapplied overhead [entry (14) above]	5,000
Adjusted cost of goods sold	$123,500

EXHIBIT 3–14
A General Model of Cost Flows



Review Problem: Job-Order Costing

Hogle Corporation is a manufacturer that uses job-order costing. On January 1, the beginning of its fiscal year, the company's inventory balances were as follows:

Raw materials	$20,000
Work in process	$15,000
Finished goods	$30,000

The company applies overhead cost to jobs on the basis of machine-hours worked. For the current year, the company estimated that it would work 75,000 machine-hours and incur $450,000 in manufacturing overhead cost. The following transactions were recorded for the year:

a. Raw materials were purchased on account, $410,000.
b. Raw materials were requisitioned for use in production, $380,000 ($360,000 direct materials and $20,000 indirect materials).
c. The following costs were accrued for employee services: direct labor, $75,000; indirect labor, $110,000; sales commissions, $90,000; and administrative salaries, $200,000.
d. Sales travel costs were $17,000.
e. Utility costs in the factory were $43,000.
f. Advertising costs were $180,000.
g. Depreciation was recorded for the year, $350,000 (80% relates to factory operations, and 20% relates to selling and administrative activities).
h. Insurance expired during the year, $10,000 (70% relates to factory operations, and the remaining 30% relates to selling and administrative activities).
i. Manufacturing overhead was applied to production. Due to greater than expected demand for its products, the company worked 80,000 machine-hours during the year.
j. Goods costing $900,000 to manufacture according to their job cost sheets were completed during the year.
k. Goods were sold on account to customers during the year for a total of $1,500,000. The goods cost $870,000 to manufacture according to their job cost sheets.

Required:

1. Prepare journal entries to record the preceding transactions.

Solution to Review Problem

1.	*a.*	Raw Materials	410,000	
		Accounts Payable		410,000
	b.	Work in Process	360,000	
		Manufacturing Overhead	20,000	
		Raw Materials		380,000
	c.	Work in Process	75,000	
		Manufacturing Overhead	110,000	
		Sales Commissions Expense	90,000	
		Administrative Salaries Expense	200,000	
		Salaries and Wages Payable		475,000
	d.	Sales Travel Expense	17,000	
		Accounts Payable		17,000
	e.	Manufacturing Overhead	43,000	
		Accounts Payable		43,000
	f.	Advertising Expense	180,000	
		Accounts Payable		180,000
	g.	Manufacturing Overhead	280,000	
		Depreciation Expense	70,000	
		Accumulated Depreciation		350,000
	h.	Manufacturing Overhead	7,000	
		Insurance Expense	3,000	
		Prepaid Insurance		10,000
	i.	The predetermined overhead rate for the year is computed as follows:		

$$\text{Predetermined overhead rate} = \frac{\text{Estimated total manufacturing overhead cost}}{\text{Estimated total amount of the allocation base}}$$

$$= \frac{\$450{,}000}{75{,}000 \text{ machine-hours}}$$

$$= \$6 \text{ per machine-hour}$$

Based on the 80,000 machine-hours actually worked during the year, the company applied $480,000 in overhead cost to production: $6 per machine-hour × 80,000 machine-hours = $480,000. The following entry records this application of overhead cost:

	Work in Process	480,000	
	Manufacturing Overhead		480,000
j.	Finished Goods	900,000	
	Work in Process		900,000
k.	Accounts Receivable	1,500,000	
	Sales		1,500,000
	Cost of Goods Sold	870,000	
	Finished Goods		870,000

Thank you

Cost-Volume-Profit (CVP) Analysis

Cost-Volume-Profit (CVP) Relationships

Understanding the interplay between costs, volume, and profit

- **Introduction to CVP Analysis**
  - **Definition:**
    CVP analysis examines how profits are affected by:
    - Selling prices
    - Sales volume
    - Unit variable costs
  - Total fixed costs **Purpose:** Helps managers make informed decisions about pricing, production, and sales strategies.

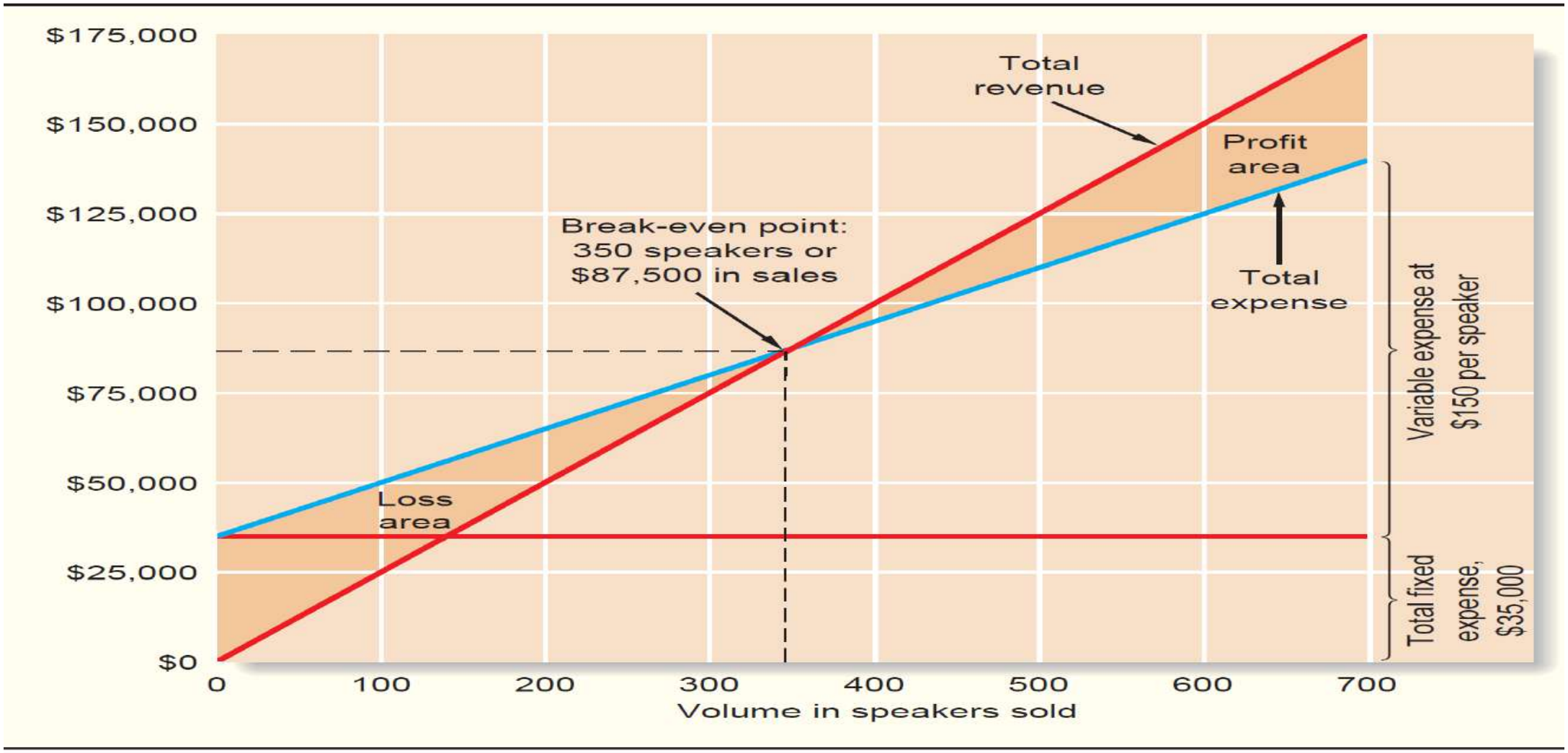
    Fixed costs are costs that do not change with the level of production or sales. They remain constant over a relevant range.

- **Key Takeaway:** Contribution margin (Sales - Variable Costs) covers fixed costs and generates profit.

Sales	$100,000	$250
Variable expenses	$60,000	$150
Contribution margin	$40,000	$100
Fixed expenses	$35,000	
Net operating income	$5,000	

- **Break-Even Point**
  - **Definition:**
    The sales level where profit is zero (Total Revenue = Total Costs).
    - **Example (350 speakers):**
    - Contribution margin = 35,000(35,000(100 × 350)
    - Fixed expenses = $35,000
    - **Net operating income = $0**
- **Beyond Break-Even:** Each additional unit sold increases profit by the unit contribution margin ($100).

- **CVP Equation**
  - **Profit = (Sales - Variable Expenses) - Fixed Expenses**
    For a single product:
    - Sales = Selling price per unit (P) × Quantity sold (Q)
    - Variable expenses = Variable cost per unit (V) × Q
    - **Profit = (P × Q - V × Q) - Fixed Expenses**
  - **Break Even**
    - **Example (351 speakers):**
      Profit = (250×351−150 × 351) - 35,000 = 100.
    - Profit = Unit CM x *Q* - Fixed expenses
    - 0 = 100 x Q – 35,000
    - Q = 35,000/100
    - Q = 350



Graph Highlights:

•**Break-even point:** 350 speakers or $87,500 in sales.

•**Profit area:** Sales > 350 speakers.

•**Loss area:** Sales < 350 speakers.

- **Contribution Margin Ratio (CM Ratio)**
  - **Formula:**

$$\text{CM Ratio} = \frac{\text{Contribution Margin}}{\text{Sales}}$$

  - CM Ratio = 40,000/100,000 = **40%**
  - Per Unit: 100/250 = **40%**

- Case Study – Voltar Company
  - **Income Statement (20,000 units):**

Sales	?	Per Unit $60	100%
Variable expenses	?	Per Unit $45	?
Contribution margin	?		?
Fixed expenses	$240,000		
Net operating income	$60,000		

Tasks:

1. Compute CM ratio and variable expense ratio.
2. Compute Break-even point
3. Assume that sales increase by $400,000 next year.

- **Summary**
  - **Key Concepts:**
  - Contribution margin drives profit after covering fixed costs.
  - Break-even analysis helps set sales targets.
  - CM ratio predicts profit impact of sales changes.
- **Application:** Use CVP to evaluate pricing, cost control, and sales strategies.

Thank you

Cost Accounting

Title: Variance Analysis in Cost Accounting
Subtitle: Based on *Managerial Accounting* by Garrison, Noreen, & Brewer
Edition: 17th Edition, McGraw-Hill (2020)

Standard Costs and Operating Performance Measures

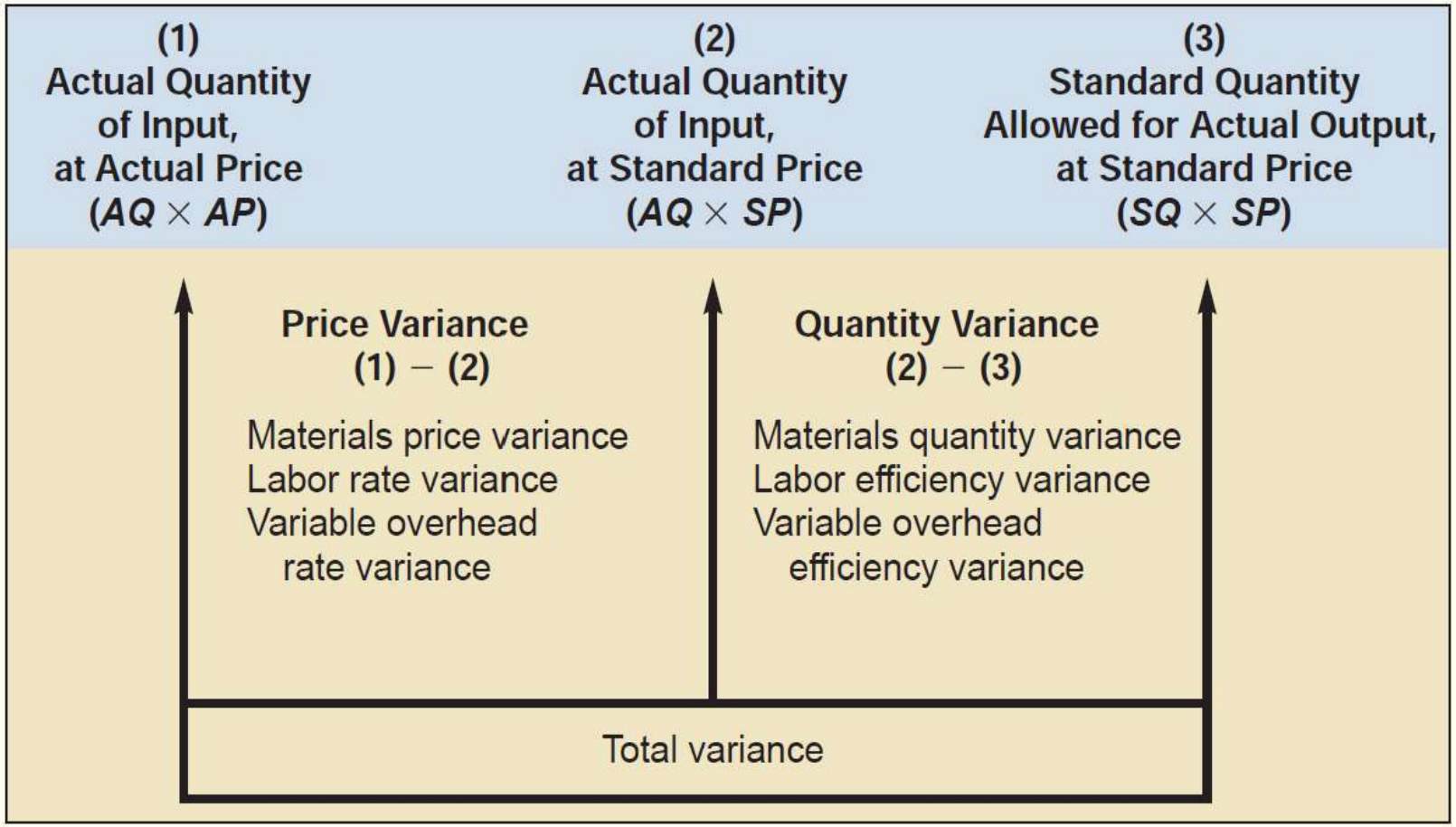
Chapter 11

- **Introduction to Variance Analysis**
  - **Key Points**:
  - **Definition**: "Comparison of actual vs. standard costs to identify deviations."
- **Purpose**:
  - Cost Control
  - Performance Evaluation
  - Budget Adjustments

Direct Material Variance Analysis

EXHIBIT 11–3

A General Model for Variance Analysis—Variable Manufacturing Costs



- **Actual Quantity (AQ)** 6,500 pounds
- **Standard Quantity (SQ)** 6,000 pounds
- **Actual Price (AP)** $3.80 per pound
- **Standard Price (SP)** $4.00 per pound

$$\text{Materials price variance} = (AQ \times AP) - (AQ \times SP)$$

Actual quantity — Actual price — Standard price

$$\text{Materials quantity variance} = (AQ \times SP) - (SQ \times SP)$$

Actual quantity — Standard price — Standard quantity allowed for actual output

EXHIBIT 11–4

Variance Analysis—Direct Materials

(1) Actual Quantity of Input, at Actual Price ($AQ \times AP$)	(2) Actual Quantity of Input, at Standard Price ($AQ \times SP$)	(3) Standard Quantity Allowed for Actual Output, at Standard Price ($SQ \times SP$)
6,500 pounds × $3.80 per pound = $24,700	6,500 pounds × $4.00 per pound = $26,000	6,000 pounds* × $4.00 per pound = $24,000

Price variance, $1,300 F

Quantity variance, $2,000 U

Total variance, $700 U

*2,000 units × 3.0 pounds per unit = 6,000 pounds.

F = Favorable; U = Unfavorable.

Labor Variance

- **Actual Hours (AH)** 5,400 hours
- **Standard Hours (SH)** 5,000 hour
- **Actual Rate (AR)** $13.75 per hour
- **Standard Rate (SR)** $14.00 per hour

$$\text{Labor rate variance} = (AH \times AR) - (AH \times SR)$$

Actual hours — Actual rate — Standard rate

$$\text{Labor efficiency variance} = (AH \times SR) - (SH \times SR)$$

Actual hours — Standard rate — Standard hours allowed for actual output

EXHIBIT 11–6

Variance Analysis—Direct Labor

Actual Hours of Input, at the Actual Rate ($AH \times AR$)	Actual Hours of Input, at the Standard Rate ($AH \times SR$)	Standard Hours Allowed for Actual Output, at the Standard Rate ($SH \times SR$)
5,400 hours × $13.75 per hour = $74,250	5,400 hours × $14.00 per hour = $75,600	5,000 hours* × $14.00 per hour = $70,000

Rate variance, $1,350 F

Efficiency variance, $5,600 U

Total variance, $4,250 U

*2,000 units × 2.5 hours per unit = 5,000 hours.

F = Favorable; U = Unfavorable.

Manufacturing Overhead

EXHIBIT 11–7

Variance Analysis—Variable Manufacturing Overhead

Actual Hours of Input, at the Actual Rate ($AH \times AR$)	Actual Hours of Input, at the Standard Rate ($AH \times SR$)	Standard Hours Allowed for Actual Output, at the Standard Rate ($SH \times SR$)
$15,390	5,400 hours × $3.00 per hour = $16,200	5,000 hours* × $3.00 per hour = $15,000

Rate variance, $810 F

Efficiency variance, $1,200 U

Total variance, $390 U

*2,000 units × 2.5 hours per unit = 5,000 hours.

F = Favorable; U = Unfavorable.

- **Actual Hours (AH)** 5,400 hours
- **Standard Hours (SH)** 5,000 hour
- **Actual Rate (AR)** $2.85 per hour
- **Standard Rate (SR)** $3.00 per hour

$$\text{Variable overhead rate variance} = (AH \times AR) - (AH \times SR)$$

Actual hours — Actual rate — Standard rate

$$\text{Variable overhead efficiency variance} = (AH \times SR) - (SH \times SR)$$

Actual hours — Standard rate — Standard hours allowed for actual output

Managerial Accounting

T h i r t e e n t h E d i t i o n
Ray H. Garrison, D.B.A., CPA
Ch-9 Budgeting

Components Of Budget

Sales Budget : Forecasts expected sales
Production Budget : Determines the number of units to be
Direct Materials Budget : Estimates the materials needed for
Direct Labor Budget : Estimates labor costs required for
Overhead Budget : Projects overhead
Cash Budget : Projects cash inflows
Budgeted Income Statement : Projects the company's income
Budgeted Balance Sheet : Projects the company's financial

1	**Total Sales**	**100,000**	**Unit, selling price per unit @ $20**
	1 Qtr	10%	
	2 Qtr	30%	
	3 Qtr	40%	
	4 Qtr	20%	

2 **Cash collection method**

* Sales are collected in the following pattern: 70% in the quarter the sales are made, and the remaining 30% in the following quarter.
* Account Receivable Beg. Balance

Sales Budget

	1 Qtr	2 Qtr	3 Qtr	4 Qtr	Year
Sales (unit)	10,000	30,000	40,000	20,000	**100,000**
Selling Price	$ 20.00	$ 20.00	$ 20.00	$ 20.00	$ 20.00
Total sales	**$200,000**	**$ 600,000**	**$800,000**	**$400,000**	**$2,000,000**

Cash Collection

Beg. Balance	$ 90,000				**$ 90,000**
1 Qtr Sales	$140,000	$ 60,000			**$ 200,000**
2 Qtr Sales		$ 420,000	$180,000		**$ 600,000**
3 Qtr Sales			$560,000	$240,000	**$ 800,000**
4 Qtr Sales				$280,000	**$ 280,000**
	$230,000	**$ 480,000**	**$740,000**	**$520,000**	**$1,970,000**

Production Budget (Unit)

Budgeted unit Sales	xxxx
Add: Desire ending Inventory	xxxx
Total Need	xxxx
less: Beg. Inv	(xxx)
required Producation	xxxx

assume

Ending inventory equal to 20% of the next quarter's sales strikes the appropriate balance.

Beg.Inv Unit 2,000

Unit	**1 Qtr**	**2 Qtr**	**3 Qtr**	**4 Qtr**	**Year**
Sales (unit)	10,000	30,000	40,000	20,000	**100,000**
Add: Desire ending Inventory	6,000	8,000	4,000	3,000	3,000
Total Producation	16,000	38,000	44,000	23,000	103,000
less: Beg. Inv	2,000	6,000	8,000	4,000	2,000
Required Producation Unit	**14,000**	**32,000**	**36,000**	**19,000**	**101,000**

Assumed

Additional Data

Each unit requires 15 pounds of Direct material

Ending Inv 10% of the following quarter's

Beg. Inv 21,000 Pound

costs $0.20 per pound,

The policy is to pay for 50% of purchases in the quarter and 50% in the following quarter,

Account Payable Beg. Balance = 25,800

Direct Material Budget

	1 Qtr	2 Qtr	3 Qtr	4 Qtr	Year
Required Producation Unit	**14,000**	**32,000**	**36,000**	**19,000**	**101,000**
Raw material need Per Unit in pound	15	15	15	15	15
Producation need	210,000	480,000	540,000	285,000	1,515,000
Add: Desire ending Inventory	48,000	54,000	28,500	22,500	22,500
Total Need	258,000	534,000	568,500	307,500	1,537,500
Less: Beg. Inv	21,000	48,000	54,000	28,500	21,000
Raw material Purchase	237,000	486,000	514,500	279,000	1,516,500
costs $0.20 per pound,	$ 0.20	$ 0.20	$ 0.20	$ 0.20	$ 0.20
Total Cost Of Raw Material	**$ 47,400**	**$ 97,200**	**$ 102,900**	**$ 55,800**	**$ 303,300**

Cash Payment Schedule

Account Payable Beg. Balance	$ 25,800				$ 25,800
1st Qrt Purchases	$ 23,700	$ 23,700			$ 47,400
2nd Qrt Purchases		$ 48,600	$ 48,600		$ 97,200
3rd Qrt Purchases			$ 51,450	$ 51,450	$ 102,900
4th Qrt Purchases				$ 27,900	$ 27,900
	$ 49,500	**$ 72,300**	**$ 100,050**	**$ 79,350**	**$ 301,200**

Direct labor budget

Addition Data

i) Each unit requires 0.40 direct labor-hour

ii) $15 per direct labor-hour

	1 Qtr	2 Qtr	3 Qtr	4 Qtr	Year
Required Producation Unit	**14,000**	**32,000**	**36,000**	**19,000**	**101,000**
Direct labour-hours per unit	0.4	0.4	0.4	0.4	0.4
Total Direct-hours need (hr)	5,600	12,800	14,400	7,600	40,400
Direct labour Cost per hour $	$15	$15	$15	$15	$15
Total Direct hours cost	**$ 84,000**	**$ 192,000**	**$ 216,000**	**$ 114,000**	**$ 606,000**

Manufacturing overhead budget

Additional data

i) Variable Manufacturing Overe Head $4 per DL-H

ii) Fixed Manufacturing OH Cost $60,600 per Qtr

iii) Depreciation on Plant & machinery $15,000 per Qtr

	1 Qtr	2 Qtr	3 Qtr	4 Qtr	Year
Total Direct-hours need (hr)	5,600	12,800	14,400	7,600	40,400
Variable Manufacturing OH Rate	$4	$4	$4	$4	$4
Variable Manufacturing OH Cost	**$ 22,400**	**$ 51,200**	**$ 57,600**	**$ 30,400**	**$ 161,600**
Fixed Manufacturing OH Cost	$ 60,600	$ 60,600	$ 60,600	$ 60,600	$ 242,400
Total Manufacturing OH Cost	**$ 83,000**	**$ 111,800**	**$ 118,200**	**$ 91,000**	**$ 404,000**
Less: Depreciation	$ 15,000	$ 15,000	$ 15,000	$ 15,000	$ 60,000
Cash Disbursement For Manf.OH	**$ 68,000**	**$ 96,800**	**$ 103,200**	**$ 76,000**	**$ 344,000**
Total Manufacturing OH Cost					**$ 404,000**
Total Direct-hours (hr)					**40,400**
Predetermine Overhead rate					**$ 10.00**

Ending finished goods inventory budget.

Item	Quantity		cost		Total
Producation cost per unit					
Direct material	15	Ponds	$ 0.20	Ponds	$ 3
Direct Labour	0.4	Hours	$ 15.00	Hours	$ 6
Manufacturing OH	0.4	Hours	$ 10.00	Hours	$ 4
Unit cost					$ 13
Budgeted Finished goods					3000
Unit cost					$ 13
Cost of FG					**$ 39,000**

Selling and administrative expense budget

Additional Data

i) Variable Selling & Admin Exp. Per unit $1.8

ii)Fixed Selling & Admin Exp.

Advertisement Exp	$	20,000	per Qtr
Exective salaries	$	55,000	per Qtr
Insurance	$	10,000	per Qtr
Property Tax	$	4,000	per Qtr
Depreciation	$	10,000	per Qtr

	1 Qtr	2 Qtr	3 Qtr	4 Qtr	Year
Budgeted sales unit	10,000	30,000	40,000	20,000	**100,000**
Variable Selling & Admin Exp. Per unit	$1.80	$1.80	$1.80	$1.80	$1.80
Variable Selling & Admin Exp.	$ 18,000	$ 54,000	$ 72,000	$ 36,000	$ 180,000
Fixed Selling & Admin Exp.					
Advertisement Exp	$ 20,000	$ 20,000	$ 20,000	$ 20,000	$ 80,000
Exective salaries	$ 55,000	$ 55,000	$ 55,000	$ 55,000	$ 220,000
Insurance	$ 10,000	$ 10,000	$ 10,000	$ 10,000	$ 40,000
Property Tax	$ 4,000	$ 4,000	$ 4,000	$ 4,000	$ 16,000
Depreciation	$ 10,000	$ 10,000	$ 10,000	$ 10,000	$ 40,000
Total Fixed Selling & Admin Exp.	$ 99,000	$ 99,000	$ 99,000	$ 99,000	$ 396,000
Total Selling & Admin Exp.	**$ 117,000**	**$ 153,000**	**$ 171,000**	**$ 135,000**	**$ 576,000**
less: Depreciation	$ 10,000	$ 10,000	$ 10,000	$ 10,000	$ 40,000
Cash Disbursement For Selling & Admin Exp.	**$ 107,000**	**$ 143,000**	**$ 161,000**	**$ 125,000**	**$ 536,000**

Cash budget

Cash Beg. Bal	xxxx
Add: All Receipts	xxxx
Total Cash Available	xxxx
Less: Disbursement	(xxx)
Excess/ Deficit	xxxx

additional data

i) Cash Beg. Balance $42,500

ii)Management plans to spend $130,000 during the year on equipment purchases: $50,000 in the fi rst quarter; $40,000 in the second quarter; $20,000 in the third quarter; and $20,000 in the fourth quarter

iii) dividend paid $8,000 per Qtr

iv)The board of directors has approved cash dividends of $8,000 per quarter.

vi)Company will have to borrow $130,000 in first Qtr, $70,000 in 2nd Qtr

	1 Qtr	2 Qtr	3 Qtr	4 Qtr	Year
Cash Beg. Balance	$ 42,500	$ 36,000	$ 33,900	$ 165,650	$ 42,500
All Receipts					$ -
Collection from Customer	$ 230,000	$ 480,000	$ 740,000	$ 520,000	$ 1,970,000
Cash Available	**$ 272,500**	**$ 516,000**	**$ 773,900**	**$ 685,650**	**$ 2,012,500**
Less: Cash Disburstment					$ -
Direct material	$ 49,500	$ 72,300	$ 100,050	$ 79,350	$ 301,200
Direct labour	$ 84,000	$ 192,000	$ 216,000	$ 114,000	$ 606,000
Manufacturing OH	$ 68,000	$ 96,800	$ 103,200	$ 76,000	$ 344,000
Selling & Administration	$ 107,000	$ 143,000	$ 161,000	$ 125,000	$ 536,000
Equipment Purchase	$ 50,000	$ 40,000	$ 20,000	$ 20,000	$ 130,000
Dividends	$ 8,000	$ 8,000	$ 8,000	$ 8,000	$ 32,000
Total Disburstment	**$ 366,500**	**$ 552,100**	**$ 608,250**	**$ 422,350**	**$ 1,949,200**
Excess/ Deficit	$ (94,000)	$ (36,100)	$ 165,650	$ 263,300	$ 63,300
Financing					$ -
Borrowing	$130,000	$ 70,000			$200,000
Repayment of Loan	-			$ (200,000.00)	$ (200,000)
Interest	-			$ (21,900.00)	$ (21,900)
Cash Balance	**$36,000**	**$33,900**	**$165,650**	**$41,400**	**$ 41,400**

Income Statement

Sales		$	2,000,000
Cost of Goods Sold	100,000unit @ 13	$	1,300,000
Gross Profit		**$**	**700,000**
less: Selling & Admin Exp		$	576,000
Net Opreating profit		$	124,000
less: Interest		$	21,900
Net profit		**$**	**102,100**

Explanation of December 31, 2008, balance sheet fi gures:
(a) The ending cash balance, as projected by the cash budget in Schedule 8.
(b) Thirty percent of fourth-quarter sales, from Schedule 1 ($400,000 30% $120,000).
(c) From Schedule 3, the ending raw materials inventory will be 22,500 pounds. This material costs $0.20 per pound. Therefore, the ending inventory in dollars will be 22,500 pounds $0.20 per pound $4,500.
(d) From Schedule 6.
(e) From the December 31, 2007, balance sheet (no change).
(f) The December 31, 2007, balance sheet indicated a balance of $700,000. During 2008, $130,000 of additional equipment will be purchased (see Schedule 8), bringing the December 31, 2008, balance to $830,000.
(g) The December 31, 2007, balance sheet indicated a balance of $292,000. During 2008, $100,000 of depreciation will be taken ($60,000 on Schedule 5 and $40,000 on Schedule 7), bringing the December 31, 2008, balance to $392,000.
(h) One-half of the fourth-quarter raw materials purchases, from Schedule 3.
(i) From the December 31, 2007, balance sheet (no change).
(j) December 31, 2007, balance $449,900
Add net income, from Schedule 9 102,100
552,000
Deduct dividends paid, from Schedule 8 32,000
December 31, 2008, balance $520,000

Balance Sheet Dec-31 2008

Assets	Amount	Ref
Current assets:		
Cash	41400	(a)
Accounts receivable	120000	(b)
Raw materials inventory	4500	(c)
Finished goods inventory	39000	(d)
Total current assets		204900
Plant and equipment:		
Land	80000	(e)
Buildings and equipment	830000	(f)
Accumulated depreciation	-392000	(g)
Plant and equipment, net		518000
Total assets		**722900**

Liabilities and Stockholders' Equity	Amount	Ref
Current liabilities:		
Accounts payable (raw materials)	27900	(h)
Total Liability		27900
Stockholders' equity:		
Common stock, no par	175000	(i)
Retained earnings	520000	(j)
Total stockholders' equity		695000
Total liabilities and stockholders' equity		**722900**

Balance Sheet Dec-2007

Assets	Amount	
Current assets:		
Cash	42500	
Accounts receivable	90000	
Raw materials inventc	4200	
Finished goods invent	26000	
Total current assets		**162700**
Plant and equipment:		
Land	80000	
Buildings and equipm	700000	
Accumulated depreci	-292000	
Plant and equipment, net		488000
Total assets		**650700**

Liabilities and Stockholders'	Amount	
Current liabilities:		
Accounts payable (rav	25800	
Total Current assets		25800
Stockholders' equity:		
Common stock, no pa	175000	
Retained earnings	449900	
Total stockholders' equity		624900
Total liabilities and stockholders'		**650700**